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Measurable Functions

1.1 Elementary Properties of Measurable Functions

Definition 1.1.1

We endow $\overline{\mathbb{R}}$ with a topology called the order topology in the following manner. For each $a \in \mathbb{R}$ let

$$L_a = \overline{\mathbb{R}} \cap \{x : x < a\} = [-\infty, a) \quad \text{and} \quad R_a = \overline{\mathbb{R}} \cap \{x : x > a\} = (a, \infty].$$

The collection $\mathcal{S} = \{L_a : a \in \mathbb{R}\} \cup \{R_a : a \in \mathbb{R}\}$ is taken as a subbasis for this topology. A basis for the topology is given by

$$\mathcal{S} \cup \{R_a \cap L_b : a, b \in \mathbb{R}, a < b\}.$$

Observe that the topology on $\mathbb{R}$ induced by the order topology on $\overline{\mathbb{R}}$ is precisely the usual topology on $\mathbb{R}$.

Definition 1.1.2 ▶ Measurable Functions

- Suppose $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ are measure spaces. A mapping $f : X \rightarrow Y$ is called **measurable with respect to $\mathcal{M}$ and $\mathcal{N}$** if $f^{-1}(E) \in \mathcal{M}$ whenever $E \in \mathcal{N}$. If there is no danger of confusion, reference to $\mathcal{M}$ and $\mathcal{N}$ will be omitted, and we will simply use the term “measurable mapping.”
- If Y is a topological space, a restriction is placed on $\mathcal{N}$. In this case it is always assumed that $\mathcal{N}$ is the σ -algebra of Borel sets $\mathcal{B}$. Thus, in this situation, a mapping $(X, \mathcal{M}) \xrightarrow{f} (Y, \mathcal{B})$ is measurable if

$$f^{-1}(E) \in \mathcal{M} \text{ whenever } E \in \mathcal{B}.$$

- When Y is taken as $\overline{\mathbb{R}}$ (endowed with the order topology) and $(X, \mathcal{M})$ is a topological space with $\mathcal{M}$ the collection of Borel sets. Then f is called a **Borel measurable function**.

- When $X = \mathbb{R}^n$, $\mathcal{M}$ is the class of Lebesgue measurable sets and $Y = \overline{\mathbb{R}}$. Here, the measurability of f required that $f^{-1}(E)$ be Lebesgue measurable whenever $E \subset \overline{\mathbb{R}}$ is Borel, in which case f is called a **Lebesgue measurable function**. The definitions imply that E is a measurable set if and only if χ_E is a measurable function.

Theorem 1.1.3

If the mapping $(X, \mathcal{M}) \xrightarrow{f} (Y, \mathcal{B})$ is measurable, where $\mathcal{B}$ is the σ -algebra of Borel sets. Define

$$\Sigma = \{E : E \subset Y \text{ and } f^{-1}(E) \in \mathcal{M}\}.$$

Then Σ is a σ -algebra.

Proof. Easy to verify. □

Corollary 1.1.4

A continuous mapping is a Borel measurable function.

Proof. If f is continuous, Σ contains all open sets. Since Σ is a σ -algebra, Σ contains all Borel sets. □

Definition 1.1.5

If $f : X \rightarrow \mathbb{R}$, we call the sets

$$\{f > a\} := \{x \in X : f(x) > a\}$$

the **superlevel sets** of f .

Theorem 1.1.6

Let $f : X \rightarrow \overline{\mathbb{R}}$, where $(X, \mathcal{M})$ is a measure space. The following conditions are equivalent:

- (i) f is measurable.
- (ii) $\{f > a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.
- (iii) $\{f \geq a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.
- (iv) $\{f < a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.

(v) $\{f \leq a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.

Theorem 1.1.7

A function $f : X \rightarrow \overline{\mathbb{R}}$ is measurable if and only if

- (i) $f^{-1}\{-\infty\} \in \mathcal{M}$ and $f^{-1}\{\infty\} \in \mathcal{M}$ and
- (ii) $f^{-1}(a, b) \in \mathcal{M}$ for all open intervals $(a, b) \subset \mathbb{R}$.

Lemma 1.1.8

If f and g are measurable functions, then the following sets are measurable:

- (i) $X \cap \{x : f(x) > g(x)\}$,
- (ii) $X \cap \{x : f(x) \geq g(x)\}$,
- (iii) $X \cap \{x : f(x) = g(x)\}$.

Proof. If $f(x) > g(x)$, then there is a rational number r such that $f(x) > r > g(x)$. Therefore, it follows that

$$\{f > g\} = \bigcup_{r \in \mathbb{Q}} (\{f > r\} \cap \{g < r\}),$$

which is a measurable set, and easily (i) holds. (ii) is just the complements of (i) with f and g interchanged. (iii) is just the intersection two of two measurable sets of type (ii), and so it too is measurable. $\square$

Definition 1.1.9

If f and g are real-valued measurable functions, then $f + g$ is undefined at points where it would be of the form $\infty - \infty$. This difficulty is overcome if we define

$$(f + g)(x) := \begin{cases} f(x) + g(x), & x \in X - B, \\ \alpha, & x \in B, \end{cases}$$

where $\alpha \in \overline{\mathbb{R}}$ is chosen arbitrarily and where

$$B := (f^{-1}\{\infty\} \cap g^{-1}\{-\infty\}) \cup (f^{-1}\{-\infty\} \cap g^{-1}\{\infty\}).$$

Theorem 1.1.10

If $f, g : X \rightarrow \overline{\mathbb{R}}$ are measurable functions, then $f + g$ and fg are measurable.

Proof. We will treat the case that f and g have values in $\mathbb{R}$. The proof is similar in the general case and is left as an exercise (see Exercise 2, Section 5.1).

To prove that the sum is measurable, define $F : X \rightarrow \mathbb{R} \times \mathbb{R}$ by

$$F(x) = (f(x), g(x))$$

and $G : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ by

$$G(x, y) = x + y.$$

Then $G \circ F(x) = f(x) + g(x)$, so it suffices to show that $G \circ F$ is measurable. Referring to Theorem 1.1.7, we need only show that $(G \circ F)^{-1}(J) \in \mathcal{M}$ whenever $J \subset \mathbb{R}$ is an open interval. Now $U := G^{-1}(J)$ is an open set in $\mathbb{R}^2$, since G is continuous. Furthermore, U is the union of a countable family, $\mathcal{F}$, of 2-dimensional intervals I of the form $I = I_1 \times I_2$, where I_1 and I_2 are open intervals in $\mathbb{R}$. Since

$$F^{-1}(I) = f^{-1}(I_1) \cap g^{-1}(I_2),$$

we have

$$F^{-1}(U) = F^{-1}\left(\bigcup_{I \in \mathcal{F}} I\right) = \bigcup_{I \in \mathcal{F}} F^{-1}(I),$$

which is a measurable set. Thus $G \circ F$ is measurable, since

$$(G \circ F)^{-1}(J) = F^{-1}(U).$$

The product is measurable by essentially the same proof. □

Definition 1.1.11 ▶ Cantor Function

Recall that the Cantor set C can be expressed as

$$C = \bigcap_{j=1}^{\infty} C_j$$

Where C_j is the union of the 2^j closed intervals that remain after the j th step of the construction. The set

$$D_j = [0, 1] - C_j$$

consists of the 2^{j-1} open intervals that are deleted at the j th step. Let these intervals be denoted by $I_{j,k}$, $k = 1, 2, \dots, 2^{j-1}$, and order them in the obvious way from left to right. Now define a continuous function f_j on $[0, 1]$ by

$$\begin{aligned} f_j(0) &= 0, \\ f_j(1) &= 1, \\ f_j(x) &= \frac{k}{2^j}, \quad x \in I_{j,k} \end{aligned}$$

and define f_j linearly on each interval of C_j . The function f_j is continuous and nondecreasing, and it satisfies

$$|f_j(x) - f_{j+1}(x)| < \frac{1}{2^j}, \quad x \in [0, 1]$$

Since

$$|f_j - f_{j+m}| < \sum_{i=j}^{j+m-1} \frac{1}{2^i} < \frac{1}{2^{j-1}}$$

it follows that the sequence $\{f_j\}$ converges uniformly to a continuous function f , called **Cantor-Lebesgue function**.

Note.

f_j 在 $I_{j,k}$ 上的设定是端点的平均值.

Corollary 1.1.12

The Cantor-Lebesgue function f and its associate $h(x) := f(x) + x$ described above have the following properties:

- (i) $f(C) = [0, 1]$; that is, f maps a set of measure 0 onto a set of positive measure.
- (ii) h maps a Lebesgue measurable set onto a nonmeasurable set.
- (iii) The composition of Lebesgue measurable functions need not be Lebesgue measurable.

(iv) Lebesgue measurable function does not preserve Borel sets.

Proof. 1. $f : [0, 1] \rightarrow [0, 1]$ is onto. It is easy to see that $f(C) = [0, 1]$, because $f(C)$ is compact and $f([0, 1] - C)$ is countable.

2.

Proof sketch.

把 C 看成是集合 $[0, 1]$ 测度为零的“骨”. $[0, 1] \setminus C$ 是 $[0, 1]$ 的测度为 1 的“肉”. f 的构造是对“骨”的展开: $f(C) = [0, 1]$; 同时对“肉”的压缩: 令 f 在 $I_{j,k}$ 上取常值. 通过令 $h : [0, 1] \rightarrow [0, 2]$. $h(x) = f(x) + x$, 我们变换映射的尺度, 不再将“肉”压缩, 而是忠实地搬运, 将每个 $I_{j,k}$ 映到大小相同的一个区间. 从而间接的看到这两个操作对“骨”的影响区别不大 (测度都是 1).

Note.

$h(C)$ 和 $f(C)$ 测度相同在直觉上也是存在依据的.

Cantor 函数 f 是一列分段线性函数 f_n 的一致极限. 在构成 Cantor 集第 n 步近似 C_n 的那些区间上, f_n 的斜率为 $(3/2)^n$, 这是一个趋向于无穷大的值.

函数 $h(x) = f(x) + x$ 可以看作是 $h_n(x) = f_n(x) + x$ 的极限. 在同样的区间上, h_n 的斜率变为 $(3/2)^n + 1$.

为一个趋于无穷的量加上 1, 并不会显著改变它的数量级. 两个斜率的比值为 $\frac{(3/2)^{n+1}}{(3/2)^n} = 1 + (2/3)^n \rightarrow 1$.

这便解释了我们的直觉: “+x” 这个操作虽然修复了 f 对 C 补集测度的“压缩”, 但对于 f 自身在 C 上的“测度生成”效应来说, 其 * 相对 * 影响是微不足道的. 因此, f 和 h 都将测度为零的集合 C “拉伸”成了一个测度为 1 的集合.

h is strictly increasing. Thus, h is a homeomorphism from $[0, 1]$ onto $[0, 2]$. Then h carries the open set $[0, 1] \setminus C$ to an open set. And we have

$$\begin{aligned} m(h([0, 1] \setminus C)) &= m\left(h\left(\bigcup_{j,k} I_{j,k}\right)\right) \\ &= \sum_{j,k} m(h(I_{j,k})) \\ &= \sum_{j,k} m(I_{j,k}) \\ &= 1 \end{aligned}$$

Therefore, h maps the Cantor set to a set P with measure 1. From Exercise ??, there exists a non-measurable set $N \subseteq P$. Set $A = h^{-1}(N)$. Since $h^{-1}(N) \subseteq h^{-1}(P) = C$, A is a subset of measure-zero set C . Thus A is Lebesgue-measurable. But $h(A) = N$ is not measurable.

3. Note that h^{-1} is measurable, since it is continuous. Let $F := h^{-1}$. Observe that χ_A is measurable, but

$$\chi_A \circ F = \chi_A \circ h^{-1} = \chi_N$$

is not Lebesgue-measurable.

4. Observe that A is not a Borel set, for if it were, then $F^{-1}(A)$ would be a Borel set. But $F^{-1}(A) = h(A) = N$ and N is not a Borel set. Now $g := \chi_A$ is a Lebesgue measurable function such that $g^{-1}(\{1\}) = A$, where $\{1\}$ is a Borel set but A is not.

□

Theorem 1.1.13

Suppose $f : X \rightarrow \overline{\mathbb{R}}$ is measurable and $g : \overline{\mathbb{R}} \rightarrow \overline{\mathbb{R}}$ is Borel measurable. Then $g \circ f$ is measurable. In particular, if $X = \mathbb{R}^n$ and f is Lebesgue measurable, then $g \circ f$ is Lebesgue measurable.

Corollary 1.1.14

Let $f : X \rightarrow \overline{\mathbb{R}}$ be a measurable function.

- (i) Let $\varphi(x) = |f(x)|^p$, $0 < p < \infty$, and let φ assume arbitrary extended values on the sets $f^{-1}(\infty)$ and $f^{-1}(-\infty)$. Then φ is measurable.
- (ii) Let $\varphi(x) = \frac{1}{f(x)}$, and let φ assume arbitrary extended values on the sets $f^{-1}(0)$, $f^{-1}(\infty)$ and $f^{-1}(-\infty)$. Then φ is measurable.

In particular, if $X = \mathbb{R}^n$ and f is Lebesgue measurable, then φ is Lebesgue measurable in (i) and (ii).

Proof. 1. Define $g(t) = |t|^p$ for $t \in \mathbb{R}$, and assign arbitrary values to $g(\infty)$ and $g(-\infty)$. Now apply the previous theorem.

2. Proceed in a similar way by defining $g(t) = \frac{1}{t}$ when $t \neq 0, \infty, -\infty$ and assigning arbitrary values to $g(0)$, $g(\infty)$, and $g(-\infty)$.

□

Theorem 1.1.15

Let $(X, \mathcal{M}, \mu)$ be a complete measure space and let f, g be extended real-valued functions defined on X . If f is measurable and $f = g$ almost everywhere, then g is measurable.

Remark.

In a complete measure space, we can consider a function f that is just defined on almost everywhere on X , by extending f into $\bar{f}$ with arbitrary value taken on where f was not defined. .

Proof. Let $N = \{x : f(x) \neq g(x)\}$. Then $\mu(N) = 0$, and thus N as well as all subsets of N are measurable. For $a \in \mathbb{R}$, we have

$$\begin{aligned} \{g > a\} &= (\{g > a\} \cap N) \cup (\{g > a\} \cap N^c) \\ &= (\{f > a\} \cap N) \cup (\{f > a\} \cap N^c) \in \mathcal{M}. \end{aligned}$$

□

Theorem 1.1.16

Suppose φ is an outer measure on a space X . Then an extended real-valued function f on X is φ -measurable if and only if

$$\varphi(A) \geq \varphi(A \cap \{f \leq a\}) + \varphi(A \cap \{f \geq b\})$$

whenever $A \subset X$ and $a < b$ are real numbers.

Proof. The “only if” is immediate from

$$\varphi(A) \geq \varphi(A \cap \{f \leq a\}) + \varphi(A \cap \{f > a\}) \geq \varphi(A \cap \{f \leq a\}) + \varphi(A \cap \{f \geq b\})$$

To show the converse, we need to prove that for any $r \in \mathbb{R}$, the set

$$E := \{f \leq r\}$$

satisfies

$$\varphi(A) \geq \varphi(A \cap E) + \varphi(A - E)$$

for each $A \subseteq X$. We define

$$B_j := A \cap \left\{ r + \frac{1}{j+1} \leq f \leq r + \frac{1}{j} \right\}$$

We shall show

$$\infty > \varphi(A) \geq \varphi\left(\bigcup_{j=1}^{\infty} B_{2j}\right) = \sum_{j=1}^{\infty} \varphi(B_{2j})$$

by induction. We fix k and assume

$$\varphi\left(\bigcup_{j=1}^k B_{2j}\right) = \sum_{j=1}^k \varphi(B_{2j})$$

We set

$$A_k := \bigcup_{j=1}^k B_{2j}$$

Then

$$\begin{aligned} (A_k \cup B_{2(k+1)}) \cap \left\{f \leq r + \frac{1}{2(k+1)}\right\} &= B_{2(k+1)} \\ (A_k \cup B_{2(k+1)}) \cap \left\{f \geq r + \frac{1}{2k+1}\right\} &= A_k \end{aligned}$$

We have

$$\varphi\left(\bigcup_{j=1}^{k+1} B_{2j}\right) = \varphi(A_k \cup B_{2(k+1)}) \geq \varphi(B_{2(k+1)}) + \varphi(A_k) = \sum_{j=1}^{k+1} \varphi(B_{2j})$$

Thus we inductively show that

$$\varphi(A) \geq \varphi\left(\bigcup_{j=1}^{\infty} B_{2j}\right) = \sum_{j=1}^{\infty} \varphi(B_{2j})$$

Similarly, one can show that

$$\varphi(A) \geq \varphi\left(\bigcup_{j=1}^{\infty} B_{2j-1}\right) = \sum_{j=1}^{\infty} \varphi(B_{2j-1})$$

Then the series

$$\sum_{j=1}^{\infty} \varphi(B_j) \leq 2\varphi(A) < \infty$$

converges. The tail of the series can be arbitrarily small. For each $\varepsilon > 0$, there exists $m \in \mathbb{N}$ such that

$$\varepsilon > \sum_{j=m}^{\infty} \varphi(B_j) \geq \varphi\left(\bigcup_{j=m}^{\infty} B_j\right)$$

Where

$$\bigcup_{j=m}^{\infty} B_j = A \cap \left\{r < f \leq r + \frac{1}{m}\right\}$$

We define a measure ψ

$$\psi(F) := \varphi(A \cap F), \quad \forall F \subseteq X$$

Then

$$\varepsilon > \psi\left(\left\{r < f \leq r + \frac{1}{m}\right\}\right)$$

Thus

$$\begin{aligned} \varphi(A \cap E) + \varphi(A - E) &= \psi(E) + \psi(E^c) \\ &= \psi(\{f \leq r\}) + \psi(\{f > r\}) \\ &\leq \psi(\{f \leq r\}) + \psi\left(\left\{f > r + \frac{1}{m}\right\}\right) + \psi\left(\left\{r < f \leq r + \frac{1}{m}\right\}\right) \\ &\leq \psi(\{f \leq r\}) + \psi\left(\left\{f \geq r + \frac{1}{m}\right\}\right) + \varepsilon \\ &= \varphi(A \cap \{f \leq r\}) + \varphi\left(A \cap \left\{f \geq r + \frac{1}{m}\right\}\right) + \varepsilon \\ &\leq \varphi(A) + \varepsilon \end{aligned}$$

Since ε is arbitrary, we have

$$\varphi(A \cap E) + \varphi(A - E) \leq \varphi(A)$$

Thus E is measurable. □

Exercises

Exercise 1.1.17

Prove that a function defined on $\mathbb{R}^n$ that is continuous everywhere except for a set of Lebesgue measure zero is a Lebesgue measurable function. In particular, conclude that a nondecreasing function defined on $[0, 1]$ is Lebesgue measurable.

Proof. If f is such function. Let N be the measure-zero set where f is not continuous. f is continuous on the topological subspace $\mathbb{R}^n \setminus N$, thus is measurable on $(\mathbb{R}^n \setminus N, \mathcal{B}(\mathbb{R}^n \setminus N))$. Then for each open sets $U \subseteq \mathbb{R}$, we have

$$\begin{aligned} f^{-1}(U) &= (f^{-1}(U) \cap (\mathbb{R}^n \setminus N)) \cup (f^{-1}(U) \cap N) \\ &= (f|_{\mathbb{R}^n \setminus N}^{-1}(U)) \cap (f^{-1}(U) \cup N) \end{aligned}$$

Where $f|_{\mathbb{R}^n \setminus N}^{-1}(U) \in \mathcal{B}(\mathbb{R}^n \setminus N)$. Note that

$$\mathcal{B}(\mathbb{R}^n \setminus N) = \{B \cap (\mathbb{R}^n \setminus N) : B \in \mathcal{B}(\mathbb{R}^n)\} \subseteq \mathcal{L}(n)$$

Since $f^{-1}(U) \cap N \in \mathcal{L}(n)$, we have $f^{-1}(U) \in \mathcal{L}(n)$. f is Lebesgue-measurable.

Since a nondecreasing function is continuous except for a countable set, which is Lebesgue-measure-zero. It is Lebesgue measurable from our argument above. $\square$

1.2 Limits of Measurable Functions

Definition 1.2.1

Let $(X, \mathcal{M}, \mu)$ be a measure space, and let $\{f_i\}$ be a sequence of measurable functions defined on X . The **upper** and **lower envelopes** of $\{f_i\}$ are defined respectively as

$$\sup_i f_i(x) = \sup\{f_i(x) : i = 1, 2, \dots\}$$

and

$$\inf_i f_i(x) = \inf\{f_i(x) : i = 1, 2, \dots\}.$$

Also, the **upper** and **lower limits** of $\{f_i\}$ are defined as

$$\limsup_{i \rightarrow \infty} f_i(x) = \inf_{j \geq 1} \left(\sup_{i \geq j} f_i(x) \right)$$

and

$$\liminf_{i \rightarrow \infty} f_i(x) = \sup_{j \geq 1} \left(\inf_{i \geq j} f_i(x) \right).$$

Theorem 1.2.2

Let $\{f_i\}$ be a sequence of measurable functions defined on the measure space $(X, \mathcal{M}, \mu)$. Then $\sup_i f_i$, $\inf_i f_i$, $\limsup_{i \rightarrow \infty} f_i$, and $\liminf_{i \rightarrow \infty} f_i$ are all measurable functions.

Proof. For each $a \in \mathbb{R}$ the identity

$$X \cap \{x : \sup_i f_i(x) > a\} = \bigcup_{i=1}^{\infty} (X \cap \{f_i(x) > a\})$$

implies that $\sup_i f_i$ is measurable. The measurability of the lower envelope follows from

$$\inf_i f_i(x) = -\sup_i (-f_i(x)).$$

Now that it has been shown that the upper and lower envelopes are measurable, it is immediate that the upper and lower limits of $\{f_i\}$ are also measurable. $\square$

Theorem 1.2.3 ▶ Egorov

Let $(X, \mathcal{M}, \mu)$ be a finite measure space and suppose $\{f_i\}$ and f are measurable functions that are finite almost everywhere on X . Also, suppose that $\{f_i\}$ converges pointwise a.e. to f . Then for each $\varepsilon > 0$ there exists a set $A \in \mathcal{M}$ such that $\mu(A^c) < \varepsilon$ and $\{f_i\} \rightarrow f$ uniformly on A .

Remark.

定义一个依赖于 $x \in X$ 和两个自然数索引 $n, k \in \mathbb{Z}_+$ 的逻辑陈述 $P_{n,k}(x)$, 它的值为 1 或 0, 代表真与假.

- k 表示精度, k 越大条件越严格.

- n 表示阶段或步骤
- 我们可以定义
- 逐点条件:

$$\forall k \in \mathbb{Z}_+, \forall x \in E, \exists N \in \mathbb{Z}_+, \forall n \geq N, P_{n,k}(x) = 1$$

即对于任意的精度 k , 在每个点上可以最终保持的.

设 $E_{n,k} = P_{n,k}^{-1}(1)$, 即该精度在某一阶段成立的点. 那么逐点条件用集合, 就是

$$E \subseteq \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} E_{n,k} = \liminf_{n \rightarrow \infty} E_{n,k}, \quad \forall k \in \mathbb{Z}_+$$

- 一致条件:

$$\forall k \in \mathbb{Z}_{>0}, \exists N_k \in \mathbb{Z}_{>0}, \forall n \geq N_k, \forall x \in E, P_{n,k}(x) = 1$$

即对于任意的精度 k , 最终总是全局保持的.

Egorov 定理就是说, 如果 $P_{n,k}(x)$ 在每个点总是最终精确的, 那么它差不多总是最终整体精确的.

Proof. Let E be the set on which the function $f_i, i = 1, 2, \dots$ are defined and finite. Let F be the set on which f convergent pointwise to f . Let $A_0 = E \cap F$. We have by hypothesis $\mu(A_0^c) = 0$. For each $\varepsilon > 0$, define

$$E_{i,k} = \left\{ x \in A_0 : |f_i - f| < \frac{1}{k} \right\}$$

Define

$$A_{i,k} = \bigcap_{j=i}^{\infty} E_{j,k} = \left\{ x \in A_0 : |f_j(x) - f(x)| < \frac{1}{k}, \forall j \geq i \right\}$$

Then $A_{1,k} \subseteq A_{2,k} \subseteq \dots, A_0 = \bigcup_{i=1}^{\infty} A_{i,k}, \forall k. A_0 \setminus A_{1,k} \supseteq A_0 \setminus A_{2,k}, \dots. \bigcap_{i=1}^{\infty} (A_0 \setminus A_{i,k}) = \emptyset.$

$$0 = \mu(\emptyset) = \lim_{i \rightarrow \infty} \mu(A_0 \setminus A_{i,k})$$

Then for each k , and for each $\varepsilon_k > 0$, there exists i_k such that

$$\mu(A_0 \setminus A_{i_k,k}) < \varepsilon_k$$

Remark.

这里就是在说, 处处最终成立的条件, 是差不多最终一直成立的.

$$E \subseteq \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} P_n \implies E_{\text{大测度, 舍弃了不多于}\varepsilon} \subseteq \bigcap_{n=N_\varepsilon}^{\infty} P_n$$

如果 P_n 都是可测条件

For each $\varepsilon > 0$, we take $\varepsilon_k = \frac{\varepsilon}{2^k}$, let $A = \bigcap_{k=1}^{\infty} A_{i_k, k}$. Then

$$\mu(A_0 \setminus A) = \mu\left(\bigcup_{k=1}^{\infty} A_0 \setminus A_{i_k, k}\right) \leq \sum_{k=1}^{\infty} \mu(A_0 \setminus A_{i_k, k}) < \sum_{k=1}^{\infty} \varepsilon_k = \varepsilon$$

$$\mu(X \setminus A) \leq \mu(X \setminus A_0) + \mu(A_0 \setminus A) = 0 + \mu(A_0 \setminus A) < \varepsilon$$

And note that for each k , since $A \subseteq A_{i_k, k}$ we have

$$|f_j(x) - f(x)| < \frac{1}{k}, \quad \forall x \in A, j \geq i_k$$

Then

$$\limsup_{i \rightarrow \infty} \sup_{x \in A} |f_i - f| = 0$$

That is $\{f_i\}$ convergents uniformly to f on A . □

Corollary 1.2.4

若存在收敛于零的正项数列 $\{a_k\}$ 使得对于

$$A_k = \{x : |f_k(x) - f(x)| \geq a_k\}$$

有级数 $\sum_{k=1}^{\infty} m(A_k)$ 收敛. 则 f_k 是近一致收敛于 f 的.

Corollary 1.2.5

Let $\{E_n\}$ be a sequence of measurable functions. If

$$E \subseteq \liminf_{n \rightarrow \infty} E_n$$

Then for each $\varepsilon > 0$, there exists measurable set E_ε , and $N_\varepsilon \in \mathbb{Z}_{>0}$, such

that $\mu(E_\varepsilon^c) < \varepsilon$ and

$$E_\varepsilon \subseteq \bigcap_{n=N_\varepsilon} E_n$$

Remark.

也就是说, 如果一系列可测的条件 P_n , 对于每个点 $x \in E$ 总是最终保持的. 相当于说每个 x 总会在每一个阶段加入并保持在 P 中, 那么我们可以让还没有加入的那些成员任意地少, 然后把它们抛弃掉, 使得剩下的成员在一个共同的阶段后总是全都保持在里面的.

Definition 1.2.6

A sequence of measurable functions $\{f_i\}$ defined relative to the measure space $(X, \mathcal{M}, \mu)$ is said to **converge in measure** to a measurable function f if for every $\varepsilon > 0$, we have

$$\lim_{i \rightarrow \infty} \mu(X \cap \{x : |f_i(x) - f(x)| \geq \varepsilon\}) = 0.$$

Theorem 1.2.7

Let $(X, \mathcal{M}, \mu)$ be a finite measure space, and suppose $\{f_i\}$ and f are measurable functions that are finite a.e. on X . If $\{f_i\}$ converges to f a.e. on X , then $\{f_i\}$ converges to f in measure.

Remark.

测度的有限性是为了保证 Egorov 的使用, 以及从一致逼近过渡到测度逼近的可行性.

Theorem 1.2.8

Let $(X, \mathcal{M}, \mu)$ be a measure space and let $\{f_i\}$ and f be measurable functions such that $f_i \rightarrow f$ in measure. Then there exists a subsequence $\{f_{i_j}\}$ such that

$$\lim_{j \rightarrow \infty} f_{i_j}(x) = f(x)$$

for μ -a.e. $x \in X$.

Remark.

依测度收敛导致子列的 a.e. 收敛.

Proof sketch.

将收敛描述成某种“逃逸难度”，依测度收敛就是在任意逃逸难度（函数值差大于 $\frac{1}{k}$ ）下，逃逸的可能性随着进程的推进最终会趋于“不可能逃逸”（逃逸出去的点最终是零测的）。于是我们选取某些进程 i_k ，使得在这个进程之后，即便逃逸难度很小，逃逸可能性也会很微弱。依测度收敛保证了几乎所有的点都被这些进程给捕获。于是最终几乎所有点在任意简单的难度下都不能逃逸，也就代表了几乎处处收敛。

Proof. Let i_1 be an integer such that

$$\mu(\{x \in X : |f_{i_1}(x) - f(x)| \geq 1\}) < \frac{1}{2}$$

Assume that we take $i_1, \dots, i_k$, then we take $i_{k+1} > i_k$ such that

$$\mu\left(\left\{x \in X : |f_{i_{k+1}}(x) - f(x)| \geq \frac{1}{k+1}\right\}\right) < \frac{1}{2^{k+1}}$$

Let

$$A_j = \bigcup_{k=j}^{\infty} \left\{x \in X : |f_{i_k} - f| \geq \frac{1}{k}\right\}$$

Then

$$A_1 \supseteq A_2 \supseteq \dots$$

$$\mu(A_j) \leq \sum_{k=j}^{\infty} \mu\left(\left\{x \in X : |f_{i_k} - f| \geq \frac{1}{k}\right\}\right) < \sum_{k=j}^{\infty} \frac{1}{2^k} = \frac{1}{2^{j-1}}$$

Specifically, $\mu(A_1) < \infty$, define $A = \bigcap_{j=1}^{\infty} A_j$ we have

$$\mu(A) = \lim_{j \rightarrow \infty} \mu(A_j) = 0$$

For each $x \in A^c = \bigcup_{j=1}^{\infty} A_j^c$, there exists k_0 such that $x \in A_{k_0}^c$,

$$A_{k_0}^c = \bigcap_{k=k_0}^{\infty} \left\{ x \in X : |f_{i_k} - f| < \frac{1}{k} \right\}$$

Which mean that

$$|f_{i_k}(x) - f(x)| < \frac{1}{k}, \quad \forall k \geq k_0$$

Thus $\lim_{k \rightarrow \infty} f_{i_k}(x) = f(x)$. Since x is arbitrary, $\{f_{i_k}\}_k$ converges to f on A^c . $\square$

Theorem 1.2.9

- A sequence $\{f_i\}$ of a.e. finite-valued measurable functions on a measure space $(X, \mathcal{M}, \mu)$ is **fundamental in measure** if for every $\varepsilon > 0$,

$$\mu(\{x : |f_i(x) - f_j(x)| \geq \varepsilon\}) \rightarrow 0$$

- Prove that if $\{f_i\}$ is fundamental in measure, then there is a measurable function f to which the sequence $\{f_i\}$ converges in measure. as i and $j \rightarrow \infty$.

Theorem 1.2.10

Let $(X, \mathcal{M}, \mu)$ be a finite measure space. Suppose that $\{f_i\}_{i=1}^{\infty}$ and f are measurable functions. Prove that $f_i \rightarrow f$ in measure if and only if each subsequence of f_i has a subsequence that converges to f μ -a.e.

1.3 Approximation of Measurable Functions

Theorem 1.3.1

Let $f : X \rightarrow \overline{\mathbb{R}}$ be an arbitrary (possibly nonmeasurable) function. Then the following hold:

- (i) There exists a sequence of simple functions, $\{f_i\}$, such that $f_i(x) \rightarrow f(x)$ for each $x \in X$.
- (ii) If f is nonnegative, the sequence can be chosen such that $f_i \uparrow f$.

- (iii) If f is bounded, the sequence can be chosen such that $f_i \rightarrow f$ uniformly on X .
- (iv) If f is measurable, the f_i can be chosen to be measurable.

Proof. Step1. All for $f \geq 0$:

Assume that $f \geq 0$. For $i \in \mathbb{Z}_{>0}$, partition $[0, i)$ into $i \cdot 2^i$ half-open intervals of the form $[\frac{k-1}{2^i}, \frac{k}{2^i})$, $k = 1, 2, \dots, i \cdot 2^i$. Label these intervals $H_{i,k}$ and let

$$A_{i,k} = f^{-1}(H_{i,k}), \quad A_i = f^{-1}([i, \infty))$$

Define f_i on X as

$$f_i(x) = \begin{cases} \frac{k-1}{2^i}, & x \in A_{i,k} \\ i, & x \in A_i \end{cases}$$

- If f is measurable, then the sets $A_{i,k}$ and A_i are measurable, and thus so are the functions f_i .

It is easy to verify that

$$f_1 \leq f_2 \leq \dots \leq f$$

If $f(x) < \infty$, then for every $i > f(x)$, we have

$$|f_i(x) - f(x)| < \frac{1}{2^i}$$

and hence $f_i(x) \rightarrow f(x)$. If $f(x) = \infty$, then $f_i(x) = i \rightarrow f(x)$. In any case, we obtain

$$\lim_{i \rightarrow \infty} f_i(x) = f(x), \quad x \in X$$

- If f is bounded by M . Then $A_i = \emptyset$ for all $i > M$, and therefore $|f_i(x) - f(x)| < \frac{1}{2^i}$ for all $x \in X$, thus showing that $f_i \rightarrow f$ uniformly.

Step 2. The general case:

Applying to f^+ and f^- , (i), (iii), (iv) holds.

□

Theorem 1.3.2 ► Lusin's theorem

Suppose $(X, \mathcal{M}, \mu)$ is a measure space where X is a metric space and μ is a finite Borel measure. Let $f : X \rightarrow \mathbb{R}$ be a measurable function that

is finite almost everywhere. Then for every $\varepsilon > 0$ there is a closed set $F \subset X$ with $\mu(X \setminus F) < \varepsilon$ such that f is continuous on F in the relative topology.

Corollary 1.3.3

Let $E \subseteq \mathbb{R}^n$ be a Lebesgue measurable set. f is a a.e. finite measurable function on E . Then for any $\delta > 0$, there exists a continuous g on $\mathbb{R}^n$ such that

$$m(\{x \in E : f(x) \neq g(x)\}) < \delta$$

If E is bounded, g can be chosen to be compactly supported.

Remark.

- If f is bounded. Then from Tiesze Extension Theorem, g can be chosen to satisfy $\sup_{x \in \mathbb{R}^n} |g| \leq \sup_{x \in F} |f|$

Proof. From Lusin's Theorem, for each $\delta > 0$, there exists a closed set F in E , with $m(E \setminus F) < \delta$ such that f is continuous on F with the relative topology. The function $f|_F$ can be extended to a continuous function g on $\mathbb{R}^n$ such that

$$f(x) = g(x), \quad \forall x \in F$$

Then it follows that

$$m(\{x \in E : f(x) \neq g(x)\}) \leq m(E \setminus F) < \delta$$

If E is bounded, assume that $E \subseteq B(0, k)$. Then from Urysohn Lemma, there is a continuous function φ with $0 \leq \varphi \leq 1$ such that

$$\varphi(F) = \{1\}, \quad \varphi((B(0, k))^c) = \{0\}$$

We substitute g by $g\varphi$, it is compactly supported. □